TRYHACKME: PS ECLIPSE

SOC Level 1 — Splunk Ransomware Investigation

Assessment type	Defensive SIEM investigation / controlled training lab
Primary tools	Splunk Enterprise, CyberChef/decoder, VirusTotal
Data sources	Windows process, file, and network/DNS telemetry
Evidence basis	Seven user-provided screenshots plus published walkthroughs
Scope	TryHackMe PS Eclipse room evidence only

Executive finding: The evidence is consistent with a ransomware intrusion chain involving an encoded PowerShell command, attempted disabling of Microsoft Defender real-time monitoring, download of `OUTSTANDING_GUTTER.exe` from an ngrok-hosted URL, scheduled-task execution as SYSTEM, outbound DNS activity, and a second-stage PowerShell script identified by VirusTotal as `BlackSun.ps1`. A ransom note and wallpaper image are also recorded as artifacts.

This report uses a Kioptrix-style finding format with separate proof-of-concept / validation sections for each major behaviour. It is an analysis of the lab's telemetry, not a claim that a live system has been compromised.

1. Executive Summary

The PS Eclipse room presents a Splunk-based ransomware investigation. The analyst searches Windows telemetry, identifies a suspicious executable in a temporary directory, decodes an encoded PowerShell command, correlates scheduled-task activity and DNS records, and uses a file hash to identify the second-stage script as BlackSun.ps1. The evidence also records a ransom note and a wallpaper image commonly associated with ransomware impact.

The most important observed indicators are the file OUTSTANDING_GUTTER.exe, PowerShell download activity from an ngrok URL, a scheduled task configured to run under NT AUTHORITY\SYSTEM, the DNS name 9030-181-215-214-32.ngrok.io, the SHA-256 hash of the PowerShell script, and BlackSun-related files.

2. Scope and Evidence

The report is limited to the TryHackMe room and supplied screenshots. URLs are defanged in this report to reduce accidental navigation. Commands and file paths are included as forensic evidence; do not run the payload or retrieve it outside an authorised isolated lab.

3. Attack Timeline (reconstructed)

Stage	Observed behaviour	Evidence / confidence
1. Initial execution	A PowerShell process runs an encoded command. Decoding reveals a command sequence that attempts to disable Defender real-time monitoring, downloads OUTSTANDING_GUTTER.exe, and creates/runs a scheduled task.	E-01 to E-03; high confidence that the command was logged, not necessarily that every action succeeded.
2. Payload download	The script uses PowerShell/Wget to download OUTSTANDING_GUTTER.exe from an ngrok-hosted address to C:\Windows\Temp.	E-02 and E-07; supported by the decoded command and challenge answers.
3. Privileged task	A task named OUTSTANDING_GUTTER.exe is created with a trigger based on Application Event ID 777 and configured with /RU SYSTEM, then started.	E-01/E-02; suspicious elevated execution/persistence configuration.
4. Remote communication	The suspicious binary is associated with a DNS query for 9030-181-215-214-32.ngrok.io.	E-04/E-07; supports attempted remote communication; does not alone prove successful C2.
5. Second stage	A script named script.ps1 is identified in C:\Windows\Temp; its SHA-256 is looked up and VirusTotal labels it BlackSun.ps1.	E-05; hash-based identification in the screenshot.
6. Ransomware impact artifacts	A ransom note and a wallpaper image are identified in the user and Public Pictures directories.	E-06/E-07; file artifacts associated with the BlackSun ransomware chain.

4. Findings — Kioptrix-Style PoC Sections

Finding 1 — Encoded PowerShell and security-control tampering

FINDING	Finding 1 — Encoded PowerShell and security-control tampering
SEVERITY	Critical (provisional)
BEHAVIOUR / TECHNIQUE	PowerShell execution / obfuscated command / impair defenses (MITRE ATT&CK; T1059.001, T1027, T1562.001)
CVE	Not assigned — evidence describes attacker behaviour and configuration, not a specific software vulnerability.
DESCRIPTION	The PowerShell command line contains a Base64-encoded string. The decoded content includes Set-MpPreference -DisableRealtimeMonitoring $true, indicating an attempt to disable Microsoft Defender real-time monitoring. The same command chain downloads a binary and creates/runs a scheduled task.
PROOF OF CONCEPT / VALIDATION	In Splunk, locate PowerShell command lines containing -enc or -EncodedCommand and preserve the original event. Decode the Base64 content in an offline analysis tool, accounting for UTF-16LE and null bytes where appropriate. Compare the decoded command with the event timestamp, user, host, and process lineage.
EVIDENCE	E-01, E-02, E-03
REMEDIATION	Enable PowerShell Script Block Logging and command-line process auditing; alert on encoded PowerShell and Set-MpPreference changes; restrict local admin rights; use tamper protection and EDR controls; investigate Defender configuration changes immediately.
LIMITATION	The event proves the command was logged. Separate telemetry is needed to verify whether Defender was actually disabled and whether all command stages succeeded.

Finding 2 — Download of OUTSTANDING_GUTTER.exe

FINDING	Finding 2 — Download of OUTSTANDING_GUTTER.exe
SEVERITY	High
BEHAVIOUR / TECHNIQUE	Ingress tool transfer / suspicious binary download (MITRE ATT&CK; T1105)
CVE	Not assigned — evidence describes attacker behaviour and configuration, not a specific software vulnerability.
DESCRIPTION	The decoded PowerShell command uses a download operation to retrieve OUTSTANDING_GUTTER.exe from an ngrok-hosted URL and save it in C:\Windows\Temp. A binary in a temporary directory, downloaded from an externally hosted tunnel, is suspicious in this context.
PROOF OF CONCEPT / VALIDATION	Search endpoint telemetry for OUTSTANDING_GUTTER.exe and review Image, CommandLine, DestinationHostname/IP, user, parent process and file-creation events. The room answer identifies the download host as hxxp[://]886e-181-215-214-32[.]ngrok[.]io. Do not browse or execute the payload; use logs and controlled analysis.
EVIDENCE	E-02, E-04, E-07
REMEDIATION	Restrict outbound web access where appropriate; block validated indicators; use EDR to quarantine the file after evidence preservation; collect its hash and signature; alert on executable downloads into Temp by PowerShell.
LIMITATION	The screenshots show the decoded command and challenge answer, but do not provide a standalone hash for OUTSTANDING_GUTTER.exe.

Finding 3 — Scheduled task configured to run as SYSTEM

FINDING	Finding 3 — Scheduled task configured to run as SYSTEM
SEVERITY	Critical (provisional)
BEHAVIOUR / TECHNIQUE	Scheduled task / privilege context / possible persistence (MITRE ATT&CK; T1053.005)

CVE	Not assigned — evidence describes attacker behaviour and configuration, not a specific software vulnerability.
DESCRIPTION	The command creates a task named OUTSTANDING_GUTTER.exe with a trigger tied to Application Event ID 777 and sets /RU SYSTEM. It then runs the task. The screenshot text shows the task action as C:\Windows\Temp\COUTSTANDING_GUTTER.exe, which appears inconsistent with the downloaded filename OUTSTANDING_GUTTER.exe. This discrepancy should be checked against the raw event rather than silently corrected.
PROOF OF CONCEPT / VALIDATION	Review the original schtasks.exe process event, task name, /TR path, /SC ONEVENT trigger, /EC Application, /MO condition, and /RU SYSTEM. Correlate with Task Scheduler Operational logs and file telemetry to verify the task was created and the referenced file ran.
EVIDENCE	E-01, E-02, E-07
REMEDIATION	Audit scheduled-task creation and changes; alert on tasks running as SYSTEM from Temp or other user-writable locations; remove unauthorized tasks after preserving XML/logs; enforce least privilege and application control.
LIMITATION	The command line is evidence of an attempt to create/run the task. The possible COUTSTANDING/OUTSTANDING path mismatch means the exact action path needs raw-event validation.

Finding 4 — Outbound DNS / possible command-and-control

FINDING	Finding 4 — Outbound DNS / possible command-and-control
SEVERITY	High (provisional)
BEHAVIOUR / TECHNIQUE	Application-layer remote communication / suspicious tunnel infrastructure
CVE	Not assigned — evidence describes attacker behaviour and configuration, not a specific software vulnerability.
DESCRIPTION	The challenge evidence associates the suspicious binary with DNS lookups for 9030-181-215-214-32.ngrok.io. The use of a tunnel-hosted domain is suspicious in the context of a downloaded binary and ransomware indicators.
PROOF OF CONCEPT / VALIDATION	Search DNS or Sysmon network telemetry for QueryName=*OUTSTANDING_GUTTER.exe* or pivot on the process and query name. Defang the indicator as hxxp[://]9030-181-215-214-32[.]ngrok[.]io for reporting. Correlate DNS answers, destination IPs, timestamps, and process identifiers.
EVIDENCE	E-04, E-07
REMEDIATION	Hunt for the domain across DNS/proxy/EDR logs; block confirmed malicious infrastructure; apply egress filtering; monitor suspicious process-to-network connections and newly observed tunnel domains.
LIMITATION	A DNS query supports attempted name resolution; it does not alone prove a successful connection, command execution by a remote operator, or data exfiltration.

Finding 5 — BlackSun ransomware script and impact artifacts

FINDING	Finding 5 — BlackSun ransomware script and impact artifacts
SEVERITY	Critical
BEHAVIOUR / TECHNIQUE	Second-stage PowerShell payload / ransomware impact (MITRE ATT&CK; T1486)
CVE	Not assigned — evidence describes attacker behaviour and configuration, not a specific software vulnerability.
DESCRIPTION	The script is initially recorded as script.ps1 in C:\Windows\Temp. The supplied VirusTotal screenshot shows SHA-256 E5429F2E44990B3D4E249C566FBF19741E671C0E40B809F87248D9EC9114BEF9 and identifies the file as BlackSun.ps1, with 31/54 vendors flagging it as malicious in that screenshot. The challenge records a ransom note and a wallpaper image path.
PROOF OF CONCEPT / VALIDATION	Hash the recovered script using SHA-256 and compare it with the observed hash. Use approved threat-intelligence sources and the original lab telemetry to confirm the identification. Search for the ransom note and wallpaper paths listed in the IOC register; preserve file metadata and event timestamps.

EVIDENCE	E-05, E-06, E-07
REMEDIATION	Isolate affected hosts when compromise is confirmed; preserve evidence; follow the ransomware response plan; identify encryption scope; restore from known-good offline backups; rotate exposed credentials where justified; block related indicators; improve EDR and backup protections.
LIMITATION	The vendor detection ratio is the value visible in the supplied screenshot and can change over time. The screenshots do not include proof of the full encryption scope or successful recovery impact.

5. IOC Register

Type	Indicator	Notes
Filename	OUTSTANDING_GUTTER.exe	Suspicious binary downloaded to C:\Windows\Temp.
Download URL	hxxp[://]886e-181-215-214-32[.]ngrok[.]io	Defanged; from decoded command and room answer.
Process path	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	Downloader process identified by challenge evidence.
Task name	OUTSTANDING_GUTTER.exe	Task configured to run as SYSTEM.
Remote DNS name	9030-181-215-214-32.ngrok.io	Associated with suspicious binary; defang before operational sharing.
Script filename	C:\Windows\Temp\script.ps1	Filename observed before hash-based identification.
Malware name	BlackSun.ps1	VirusTotal screenshot labels the file BlackSun.ps1.
SHA-256	E5429F2E44990B3D4E249C566FBF19741E671C0E40B809F87248D9EC9114BEF9	Hash shown in VirusTotal screenshot for the PowerShell script.
Ransom note	C:\Users\keegan\Downloads\vasg6b0wmw029hd\BlackSun_README.txt	Ransomware-related file artifact.
Wallpaper image	C:\Users\Public\Pictures\blacksun.jpg	Image reportedly saved to replace desktop wallpaper.

Operational note: validate every indicator in the actual environment before blocking. The ngrok hostnames are defanged in this report to reduce accidental access.

6. Splunk Queries for Reproduction

Search all events for the suspicious binary

```
index=* "OUTSTANDING_GUTTER.exe"
```

Find encoded PowerShell command lines

```
index=* powershell CommandLine="* -enc *"
```

Review scheduled task commands

```
index=* "schtasks.exe" "OUTSTANDING_GUTTER.exe"
```

Find the remote DNS name

```
index=* QueryName="9030-181-215-214-32.ngrok.io"
```

Locate PowerShell scripts and hash fields

```
index=* "script.ps1"
```

Search ransomware artifacts

```
index=* "BlackSun" OR "blacksun.jpg" OR "BlackSun_README.txt"
```

These are starter searches. Your lab's indexes and field names may differ; inspect the raw event and adjust fields such as CommandLine, Image, QueryName, TargetFilename, Hashes, EventCode, and host.

7. Mitigation and Response Plan

Contain and preserve: If this were a live incident, isolate the endpoint through the approved EDR/network process, preserve volatile and disk evidence, and export relevant Splunk events before cleanup.

Restore security controls: Confirm Microsoft Defender real-time monitoring, tamper protection, and policy enforcement are enabled. Investigate who/what changed settings and alert on future changes.

Remove persistence safely: Export the scheduled task and logs, verify its action path, then disable/remove it only after preserving evidence and confirming it is unauthorized.

Hunt for payload execution: Search for OUTSTANDING_GUTTER.exe, script.ps1, PowerShell child processes, suspicious file writes, ransom-note creation, and wallpaper modification. Identify any additional affected endpoints.

Network containment: Search proxy, DNS, firewall, and EDR telemetry for both ngrok hostnames. Block validated indicators and restrict outbound access based on organisational policy.

Ransomware recovery: Determine whether files were encrypted and assess scope. Restore from tested, known-good offline/immutable backups; do not assume removal of the note or wallpaper resolves compromise.

Detection engineering: Build alerts for encoded PowerShell, Defender disablement, certutil/PowerShell downloads into Temp, scheduled tasks running as SYSTEM from writable paths, and suspicious tunnel-domain DNS queries.

Credential and access review: Review credentials and privileged activity on the endpoint; reset credentials only when exposure or compromise warrants it, and check for lateral movement.

8. Conclusion — How the Attack Was Done

The most likely chain, as reconstructed from the lab evidence and online walkthroughs, begins with an encoded PowerShell command. After decoding, the command attempts to disable Microsoft Defender real-time monitoring, downloads OUTSTANDING_GUTTER.exe from an ngrok-hosted address into C:\Windows\Temp, and creates then runs a scheduled task configured to execute under NT AUTHORITY\SYSTEM. This combines obfuscation, defense evasion, ingress tool transfer, and privileged scheduled-task execution.

The binary is then associated with a second ngrok DNS name. A PowerShell script saved as script.ps1 is identified by its SHA-256 as BlackSun.ps1 using VirusTotal. The lab evidence then records BlackSun_README.txt as a ransom note and C:\Users\Public\Pictures\blacksun.jpg as a wallpaper image. Together these artifacts are consistent with a BlackSun ransomware incident.

The most important mitigation is early containment followed by evidence preservation, validation of whether the payload executed, restoration of security controls, removal of unauthorized scheduled tasks, and recovery from clean backups if encryption occurred. Detections should focus on the full behaviour chain rather than relying on a single filename. Note that the scheduled task's /TR path appears as COUTSTANDING_GUTTER.exe in the screenshot, so the original raw event should be checked to resolve that apparent typo.

9. Online Walkthrough References

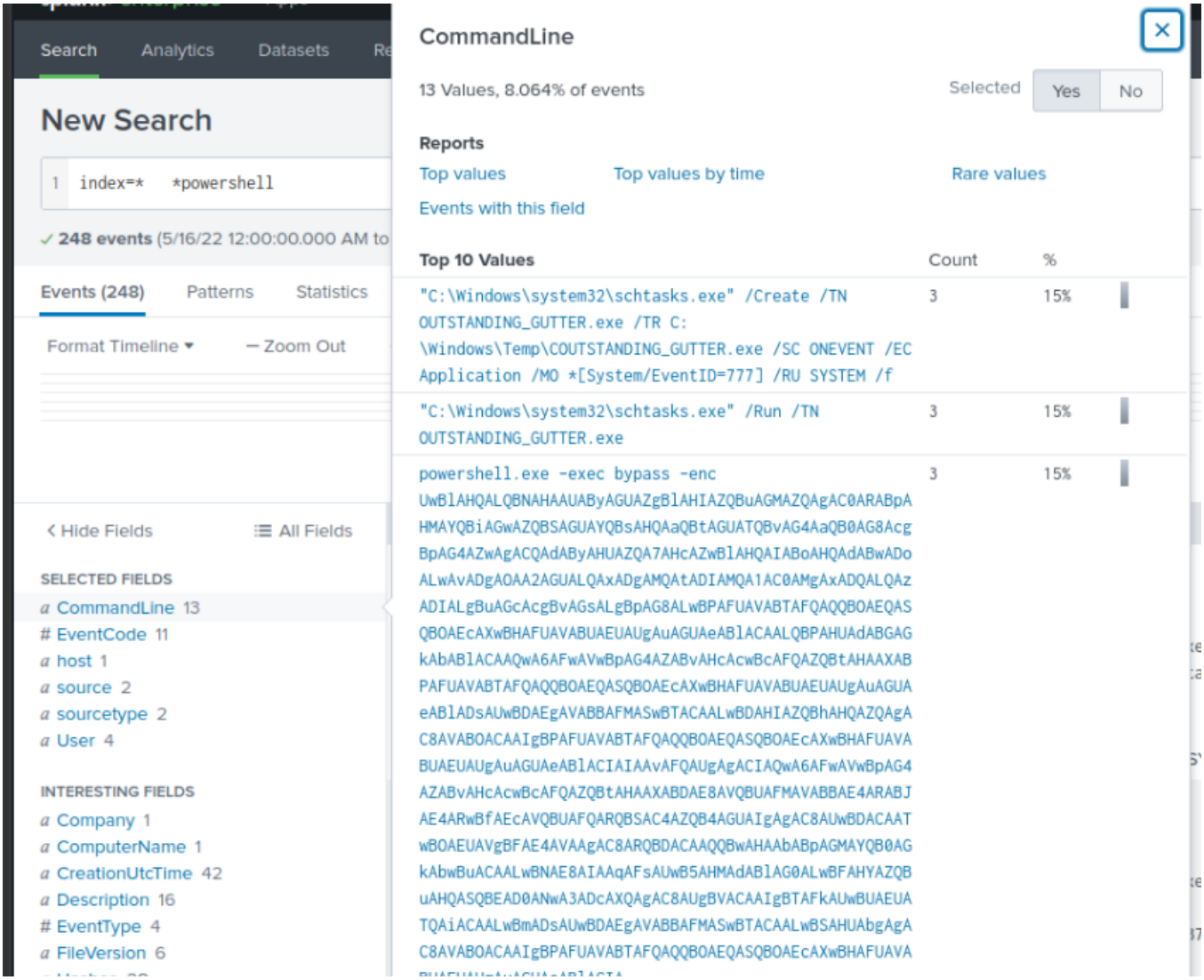
1. TryHackMe — PS Eclipse (official room; room URL may require login).
2. siunam's PS Eclipse walkthrough — Splunk investigation and challenge artifacts.
3. Ransomware Investigation with Splunk — PS Eclipse.
4. Furkanctiner — PS Eclipse walkthrough.
5. GitHub — PS Eclipse notes.

Walkthroughs are used as secondary references. Where possible, this report prioritises values visible in the user's screenshots and labels inferred behaviour as likely rather than proven.

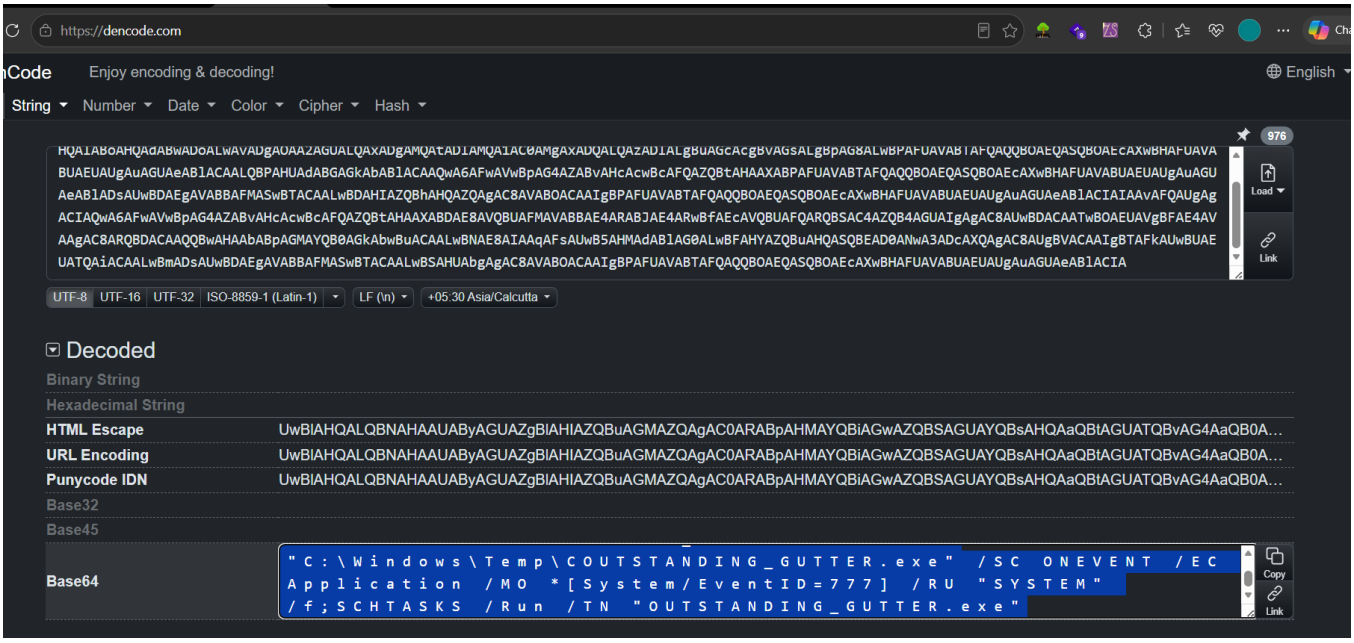
Appendix A — Screenshot Evidence Register

The screenshots below are reproduced in the same report and referenced as E-01 through E-07.

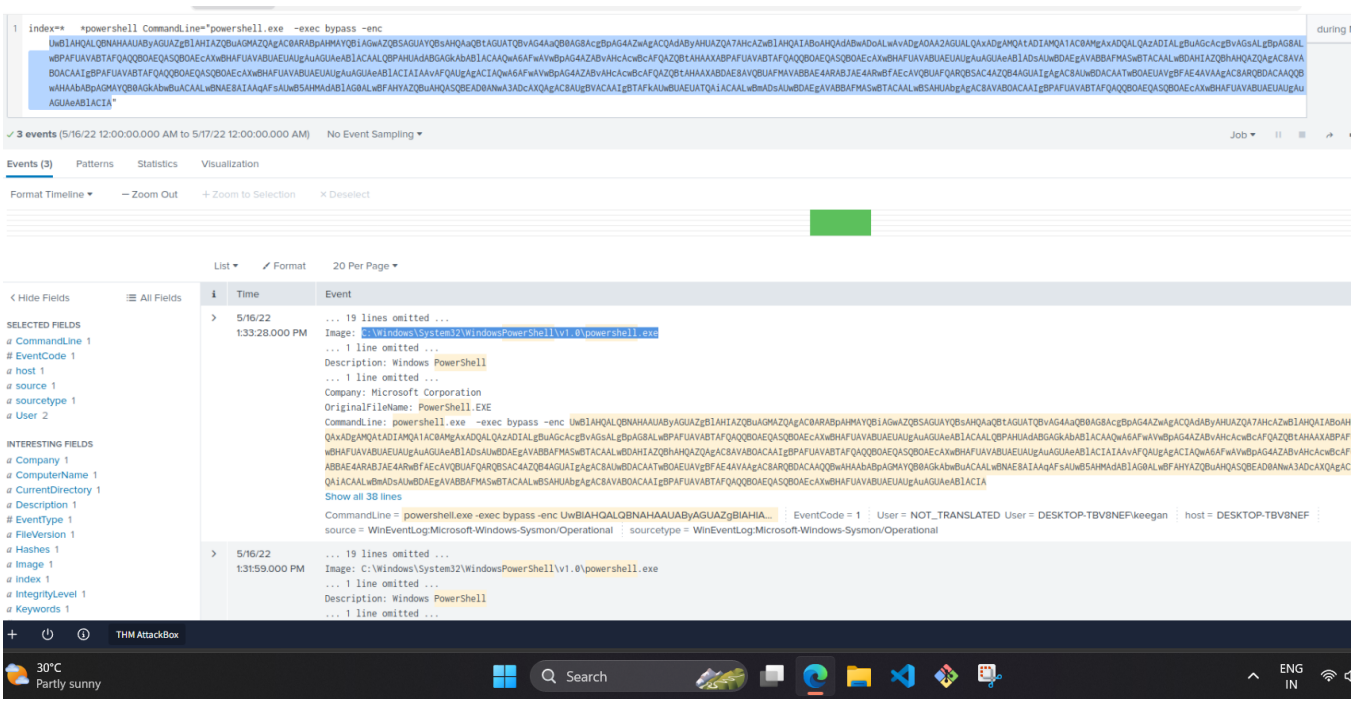
E-01 — Splunk CommandLine field shows encoded PowerShell and scheduled-task activity, including OUTSTANDING_GUTTER.exe.



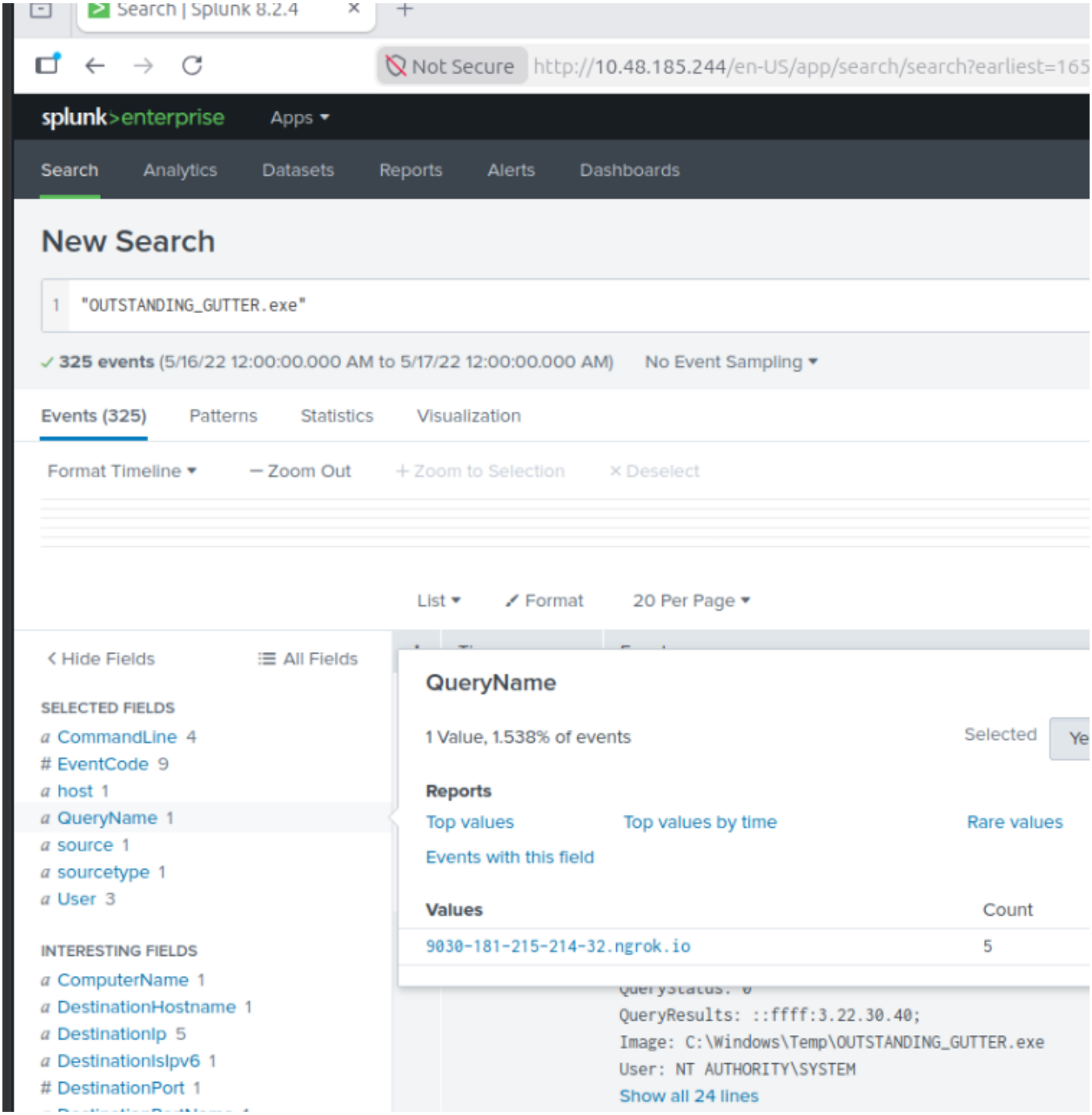
E-02 — Decoding the Base64 PowerShell command reveals Defender real-time monitoring being disabled, a download from an ngrok URL, and scheduled-task creation.



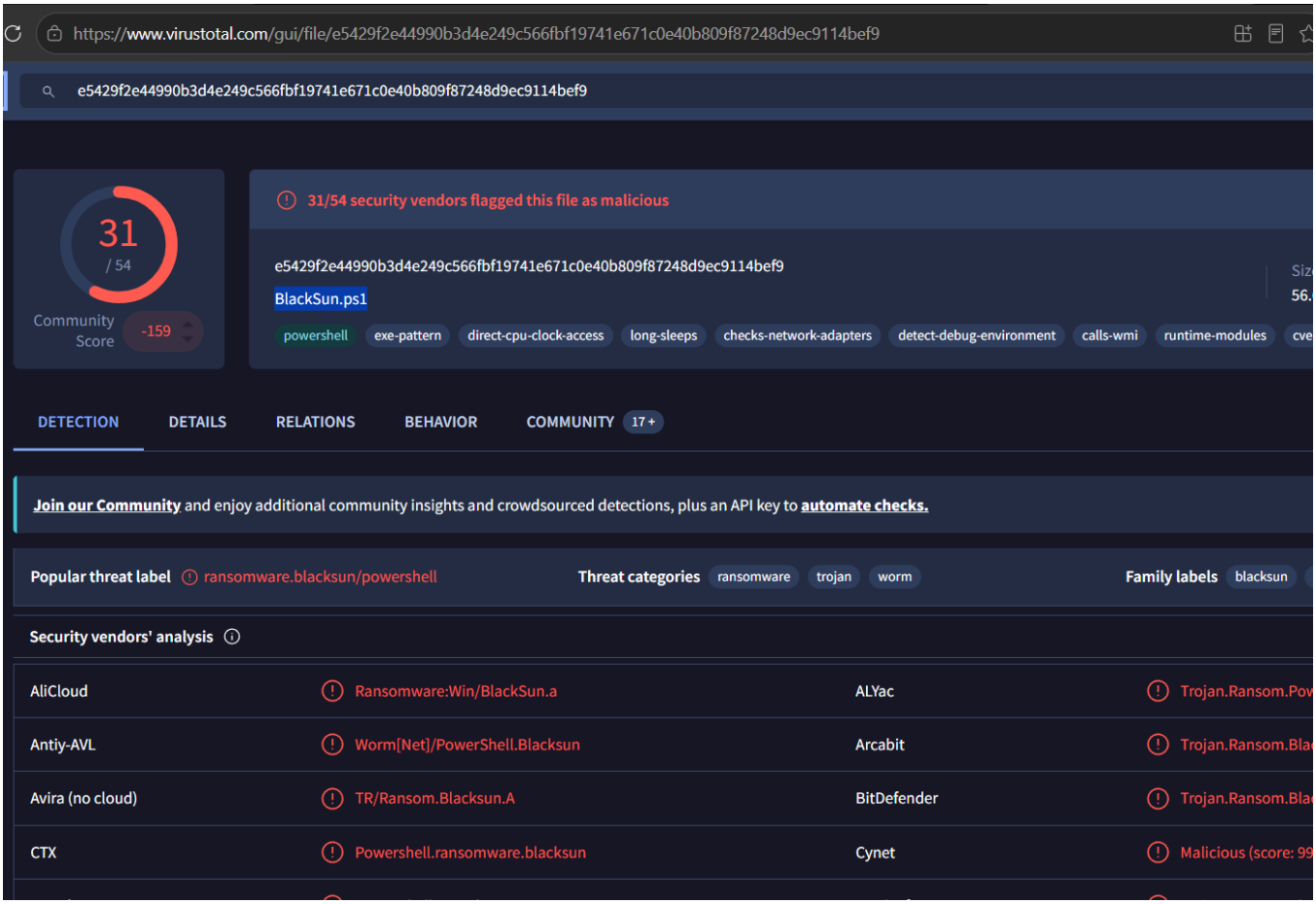
E-03 — Splunk event results show PowerShell process creation and the encoded command line.



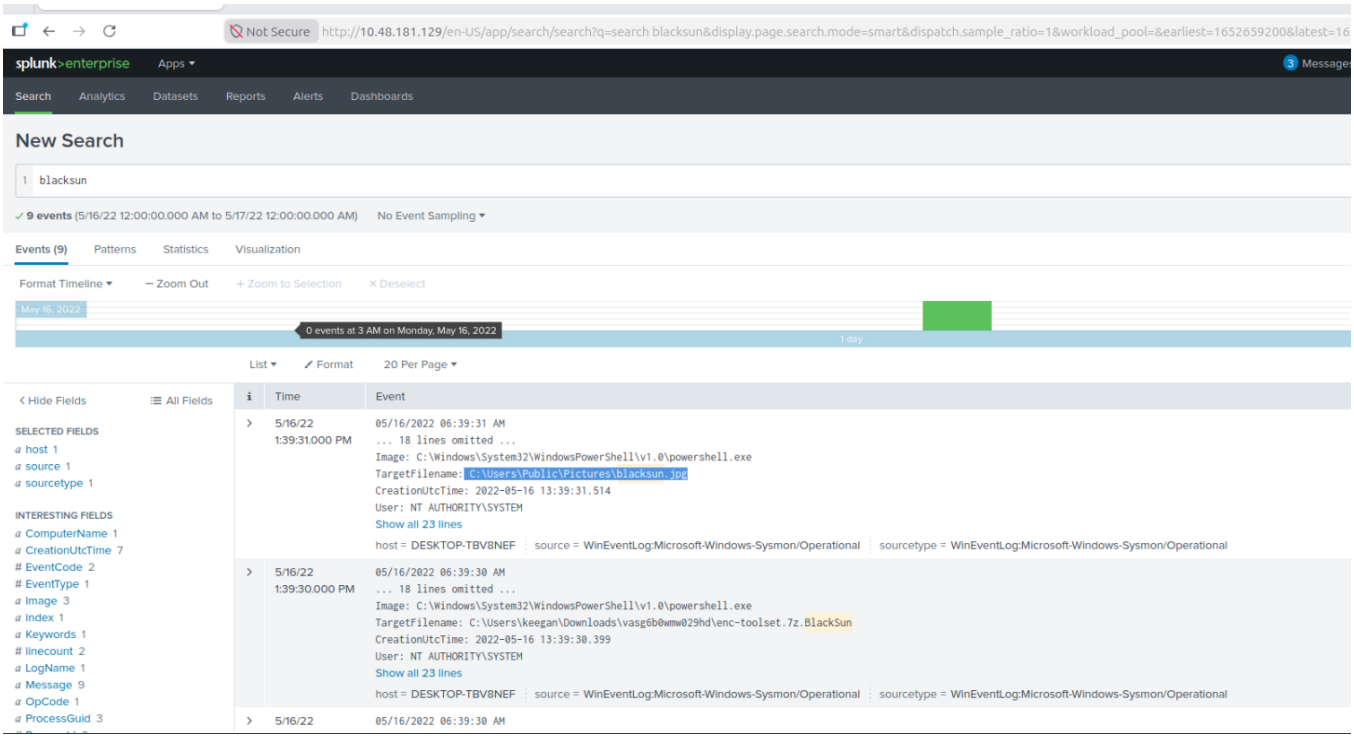
E-04 — Search results show the suspicious file OUTSTANDING_GUTTER.exe and the associated DNS query name.



E-05 — VirusTotal lookup identifies the script as BlackSun.ps1; the displayed hash is SHA-256 E5429F2E44990B3D4E249C566FBF19741E671C0E40B809F87248D9EC9114BEF9.



E-06 — Splunk results show files associated with BlackSun, including the wallpaper image path.



E-07 — TryHackMe challenge answers and IOCs recorded in the completed lab.

